



MINING DEALER BEST PRACTICE

# CRC Expert TV

Maintenance and  
Repair

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|                                               |   |
|-----------------------------------------------|---|
| CRC Expert TV .....                           | 0 |
| 1.0 Introduction .....                        | 1 |
| 2.0 Best Practice Description .....           | 1 |
| 3.0 Implementation Steps .....                | 6 |
| 4.0 Benefits .....                            | 6 |
| 5.0 Resources Required .....                  | 6 |
| 6.0 Supporting Attachments / References ..... | 7 |
| 7.0 Related Best Practices .....              | 7 |
| 8.0 Acknowledgements .....                    | 8 |

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January 2014  
1310-4.04-1324

## 1.0 Introduction

Creating a standard work is essential to control quality in workshops. Considering that there is a different level of experience among technicians, it becomes more challenging. For non-English speaking countries due to the language barrier, SIS cannot be used effectively by the technician on a daily basis on the floor.

This Best Practice presents a different approach to efficiently handle the in-place training of the technicians by using the visual (with video support) instructions together with the processes & procedures written, which helps to keep the quality of the work done and reduce the rework.

## 2.0 Best Practice Description

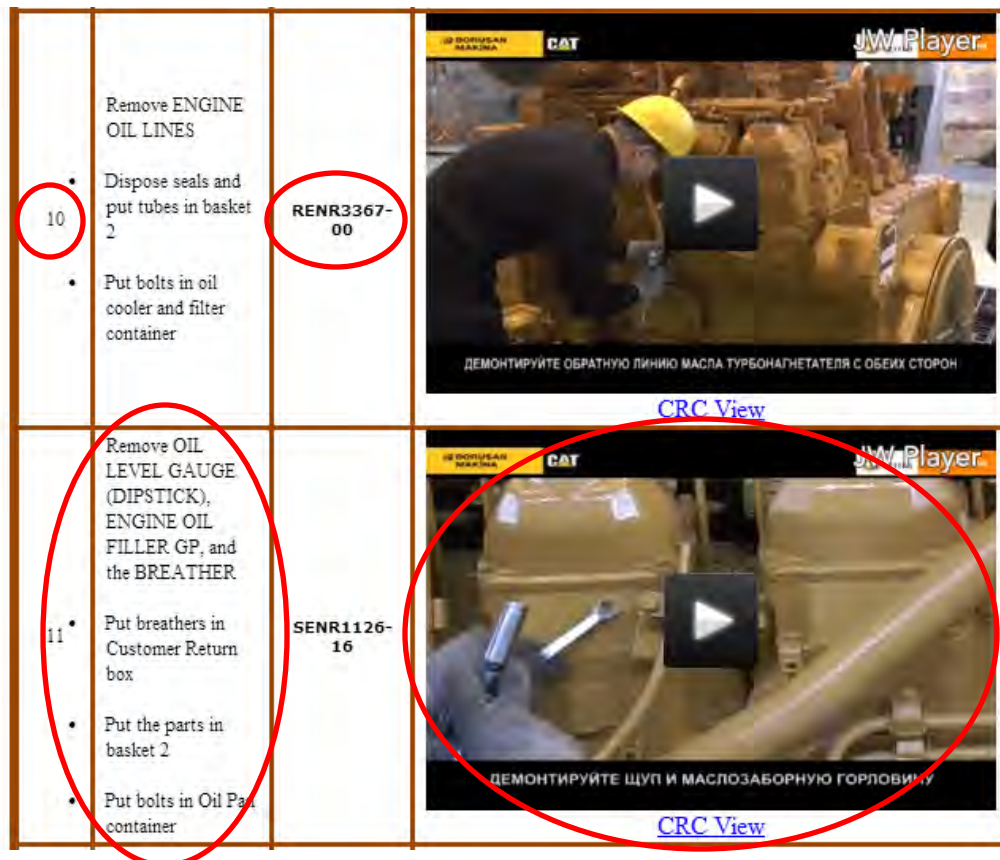
During the daily operation on the shop floor, the technicians frequently refer to SIS and other in-house developed standard work procedures.

CRC Expert TV combines SIS information (Disassembly-Assembly-Reusability Guidelines) with CRC specific processes and demonstrates it visually with video support. It represents a Step-by-step Operating Instructions/ Procedures with videos for engines and drive train components relating to Disassembly & Assembly processes. The instruction/procedure consist of:

- Step Number
- Step description
- Reference (the source of the information described)
- Video (showing how to do)
- Control Checklists

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|---------------|-------------------|-----------------|--------------------------|



Screenshot view from 3500 SERIES ENGINE DISASSEMBLY PROCEDURE

Note: In addition to the common components' disassembly/assembly processes, some key points are described in the steps and shown in the videos accordingly. These are:

- Critical areas to pay attention while preparing defect list
- How to measure parts to determine reusability
- How to perform special tests on the parts (i.e. crack testing)
- Selecting correct tools for the job and instructions to use them

For quick access and friendly use of the CRC Expert TV, the customized software was developed. Every technician has their own login and password to enter the software at any time for self-training, updating the knowledge and skills or basically for assistance while going through the component disassembly/assembly processes.

Each work bay is equipped with touch screen Kiosk terminals which make the system available all the time. Also, team leaders/Supervisors use an iPad application that allows them to check & control checklists as well as show the training video required to the technicians.

As the Expert TV is based on the software, it enables timely tracking and easy updating of the existing instructions/procedures and/or adding new ones if some changes are made and/or a new component/equipment is received by CRC.

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3. SPACER PLATE  
☐ NEW  
☐ REUSED. SPACER PLATE THICKNESS (X) FALLS WITHIN SPECS ABOVE

4. CALCULATE ACTUAL LINER PROJECTION FOR REUSED CYLINDER LINERS OR SPACER PLATES AND COMPARE TO THE SPECS IN THE ABOVE TABLE

5. INSTALL PLATE GASKET, SPACER PLATE, AND CYLINDER LINER WITHOUT SEALS AND BANDS IN THE CYLINDER BLOCK

6. INSTALL FOUR (4) BOLTS AND WASHERS THAT HOLD THE SPACER PLATES AS SHOWN IN FIGURE 1 AND TIGHTEN EVENLY IN THREE STEPS TO A TORQUE OF **70 LB-FT**

| TORQUE PER STEP, LB-FT | APPLIED TORQUE, LB-FT |
|------------------------|-----------------------|
| 25                     |                       |
| 50                     |                       |
| 70                     |                       |

7. INSTALL TOOLS TO MEASURE THE PROJECTION AS SHOWN IN FIG 1 AND TIGHTEN THE SCREWS EVENLY IN THREE STEPS TO A TORQUE OF **50 LB-FT**

| TORQUE PER STEP, LB-FT | APPLIED TORQUE, LB-FT |
|------------------------|-----------------------|
| 15                     |                       |
| 20                     |                       |
| 30                     |                       |

8. MEASURE THE PROJECTION AT FOUR POINTS

| CYLINDER | PROJECTIONS |         |         |         |
|----------|-------------|---------|---------|---------|
|          | POINT A     | POINT B | POINT C | POINT D |
| 1        |             |         |         |         |
| 2        |             |         |         |         |
| 3        |             |         |         |         |
| 4        |             |         |         |         |
| 5        |             |         |         |         |

Submit Form

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6. Clean 1 counterbore with lint free towel and isopropyl alcohol  
**Note:** Do not use pressurized air blast to clean counterbore.

7. Clean counterbore's insert with lint free towel and isopropyl alcohol.  
**Note:** Do not use pressurized air blast to clean counterbore.

8. Verify counterbore/insert are clean/dry

9. Apply Loctite 640 to counterbore bottom radius and spread to sides/bottom. Assure applicator tip is debris free.  
**Note:** Insert must be installed/checked within 5 minutes of Loctite application

10. Press insert into counterbore (part number facing up) with installation tool. Tighten press bolts in a star sequence:  
 - to 40 Nm (30 ft-lbs);  
 - to 90 Nm (65 ft-lbs);  
 - to 200 Nm (150 ft-lbs);  
 - Verify at 200 Nm (150 ft-lbs).

Screenshots how CRC Expert TV is used in iPad

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CHG  
NO  
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NUMBER  
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Screenshots how CRC Expert TV is used in a touch screen kiosk terminal by technicians

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| MAIN JOURNAL INSPECTION MEASUREMENT |                            |                                |        |              |           |       |                     |             |               |                      |              |
|-------------------------------------|----------------------------|--------------------------------|--------|--------------|-----------|-------|---------------------|-------------|---------------|----------------------|--------------|
| MAIL JOURNAL                        | MEASURED JOURNAL DIAMETERS |                                |        | OUT-OF-ROUND |           | TAPER | FILLET RADIUS CHECK | HARDNESS Rc | PROFILE CHECK | SURFACE FINISH CHECK | VISUAL CHECK |
|                                     | FRONT 1                    | CENTER 3<br>90 deg<br>TO 1 & 2 | REAR 2 | 1 MINUS 3    | 1 MINUS 2 |       |                     |             |               |                      |              |
| 1                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 2                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 3                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 4                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 5                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 6                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 7                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 8                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |
| 9                                   |                            |                                |        |              |           |       |                     |             |               |                      |              |

Technician can switch to procedure to see how to do the step in View procedure section

|                             |                                                                                                                                                                                                               |                                                                                              |                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| MAIN MEASURING LOCATION<br> | JOURNAL DISPOSITION<br><input type="checkbox"/> POLISH<br><input type="checkbox"/> GRIND<br><input type="checkbox"/> CORE RETURN<br><input type="checkbox"/> HOLD FOR _____<br><input type="checkbox"/> SCRAP | JOURNAL BENT MEASUREMENT<br>BEFORE STRAIGHTENING<br>AFTER STRAIGHTENING<br>1-2<br>2-3<br>3-4 | MISCELLANEOUS<br>MAGNETIC CRACK INSPECTION<br><input type="checkbox"/> BEFORE GRINDING<br><input type="checkbox"/> AFTER GRINDING/STRAIGHTENING |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                       |             |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--|
| 10. Remove ENGINE OIL LINES<br>Dispose seals and put tubes in basket 2<br>Put bolts in oil cooler and filter container                                                                | RENK3367-00 |  |
| 11. Remove OIL LEVEL GAUGE (DIPSTICK), ENGINE OIL FILLER OP. and the BREATHER.<br>Put breathers in Customer Return box<br>Put the parts in basket 2<br>Put bolts in Oil Pan container | SEN1126-16  |  |

Technician is able to continue filling out the checklist after page is switched back to view procedure.

Screenshot view how CRC Expert TV is used in a daily work

|                                                                                                                 |                   |              |                          |
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### 3.0 Implementation Steps

- Gather a team/committee to create/review processes including Training, HSE and I.T. departments.
- Determine the scope of the project.
- Determine/map “As Is” rebuild processes.
- Evaluate the current practice for improvements.
- Prepare the new standard work (instruction/ procedure) based on SIS, and the dealer’s own experience.
- Review “To Be” processes (instruction/procedure) with the Training Department and HSE Department in terms of safety updates. Make the necessary corrections/additions if they are required.
- Develop/improve current checklists according to new updates.
- Organize necessary training for ready and approved standard instruction/procedure with the technicians, whose work process has been chosen.
- After implementation of new processes, schedule the work area for video shooting of all the steps in the standard instruction/procedure.
- As soon as all the steps of the standard instruction/procedure are shot in video, they should be uploaded into the CRC Expert TV software, which has been created by I.T. Department for this purpose.

The Expert TV has a feedback system to contact with the technical support team. It is continuously updated according to the CRC operation needs.

### 4.0 Benefits

- Providing a good technical visual training for a new or experienced technician, which saves time and money compared to the training in class
- Increase sustainability of standard work in processes
- Maintain the high quality of the work done
- Reduce the rework percentage
- Training material for class trainings

### 5.0 Resources Required

- On average, \$600 USD per procedure to be shot by an Operator.
- Necessary tooling determined, support equipment (overhead cranes, benches, etc.).
- Technicians, Training department, Technical support department and Safety Engineer.
- IT support.

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| CRC Expert TV | DATE<br>2/19/2015 | CHG<br>NO<br>00 | NUMBER<br>1310-4.04-1324 |
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## 6.0 Supporting Attachments / References

When preparing the instructions/procedures the following information sources have been used:

- SIS (CAT standard instructions/procedures for components disassembly/assembly manuals, reusability guidelines and specifications).

EDITOR'S NOTE: Though Borusan cannot share the software code, they may be contacted if another dealer wishes to implement and has technical questions.

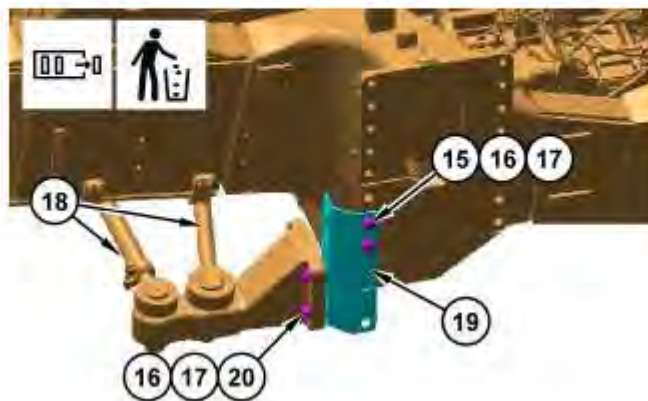
## 7.0 Related Best Practices

EDITOR'S NOTE: Caterpillar is expanding its SIS capabilities to include industry standard graphical icons to help make procedures quicker to follow. As NextGen SIS (phases 1 & 2) are rolled out in 2014-2015, expect 3D graphical animations to be made available as well. Each dealer will need to determine if they want to tailor a video strategy as defined in this Best Practice, or utilize the new features being added to SIS.

Currently you can reference media number UENR2257 (793F Off-Highway Truck Field Assembly) for examples of the new icons in use. More documents are being updated with these new graphics as time and resources allow.

1. Clean the mating surface free of wax, dirt, debris, paint, and grease.

[View Image](#)



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## 8.0 Acknowledgements

The idea of creating & implementing this Best Practice belongs to

|                                                                                                                                                                                            |                                                                                                                                                                                            |                                                                                                                                                                                              |                                                                                                                                                                                                         |
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